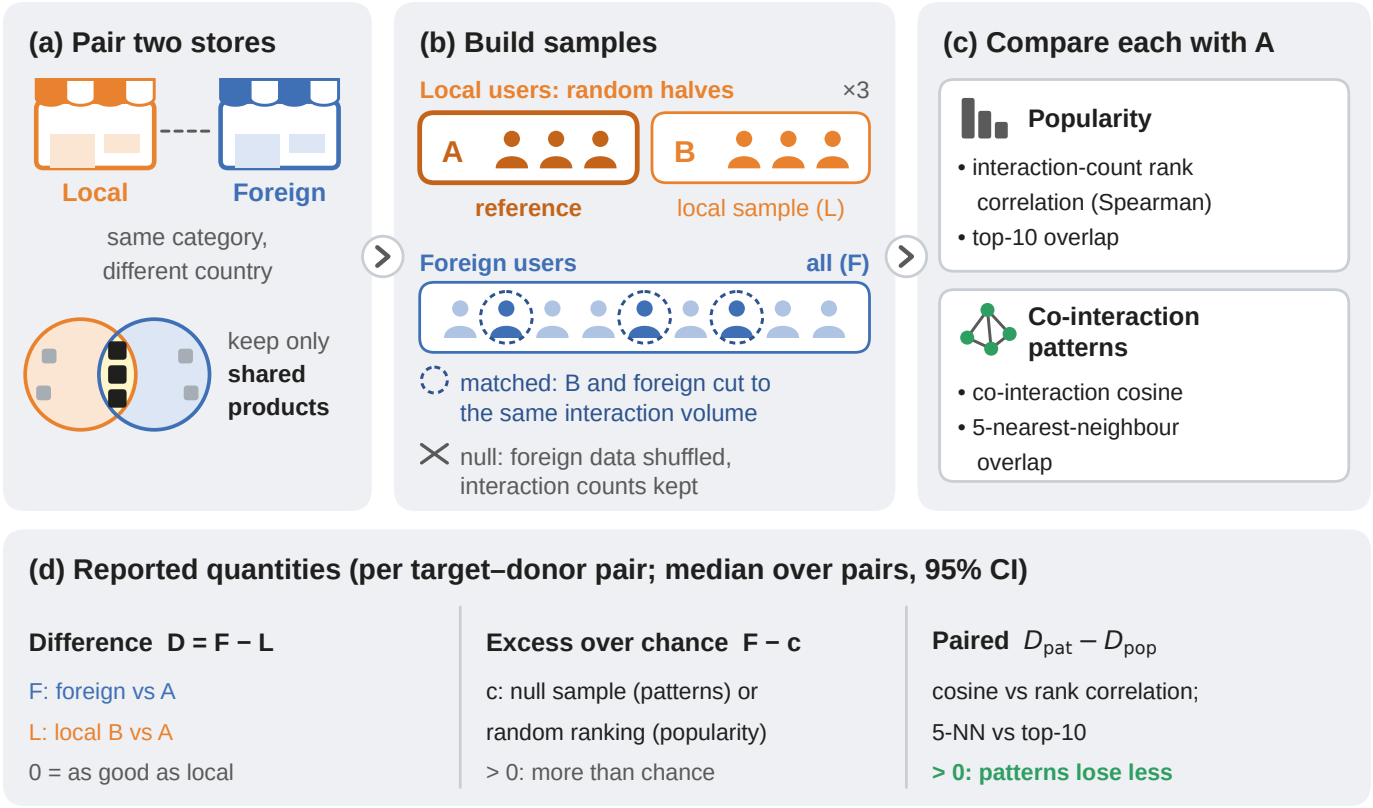
(a) Pair two stores
Local
Foreign
same category, different country
keep only shared products
(b) Build samples
Local users: random halves
×3
A
B
reference
local sample (L)
Foreign users
all (F)
matched: B and foreign cut to the same interaction volume
null: foreign data shuffled, interaction counts kept
(c) Compare each with A
Popularity
• interaction-count rank correlation (Spearman)
• top-10 overlap
Co-interaction patterns
• co-interaction cosine
• 5-nearest-neighbour overlap
(d) Reported quantities (per target–donor pair; median over pairs, 95% CI)
Difference D = F − L
F: foreign vs A
L: local B vs A
0 = as good as local
Excess over chance F − c
c: null sample (patterns) or random ranking (popularity)
> 0: more than chance
Paired $D_{pat} - D_{pop}$
cosine vs rank correlation; 5-NN vs top-10
> 0: patterns lose less